

EXCEL

COMPLETE JOB-READY LEARNING GUIDE

A practical, example-first workbook for learning Excel from the basics to workplace reporting

**FORMULAS • DATA CLEANING • LOOKUPS • DATES • PIVOTTABLES • CHARTS •
POWER QUERY • DASHBOARDS**

How this guide is different: every major topic follows the same learning pattern: What it means -> Example -> Formula -> What Excel returns -> Practice -> Common mistake.

This edition expands the uploaded The Job-Ready Excel Handbook. The original source covers data cleaning, conditional aggregation, lookups, IF statements, PivotTables, business scenarios, Power Query, validation, dates and conditional formatting. The examples and practice structure below are added to make those topics easier to learn independently.

Contents. How to use this guide

Part	Topic	Goal
1	Excel foundations	Understand the workbook and data structure
2	References & formulas	Write formulas confidently
3	Text & data cleaning	Clean real exported data
4	Conditional calculations	Answer business questions
5	Lookups	Connect two datasets
6	Dates & time	Work with deadlines and periods
7	Formatting & validation	Build controlled trackers
8	PivotTables	Summarize large datasets
9	Charts & reporting	Present results clearly
10	Power Query	Automate repeated cleaning
11	Dashboards	Build management-ready reports
12	Modern Excel	Dynamic arrays and advanced tools
13	Projects	Practice realistic workplace tasks
14	Interview preparation	Prepare for practical tests
15	30-day roadmap	Follow a structured learning plan

1. Excel Foundations

Start with the way Excel thinks about data.

1.1 Workbook, worksheet, cell and range

Term	Simple meaning	Example
Workbook	The Excel file	Sales_Report.xlsx
Worksheet	A sheet inside the file	January
Cell	One location	B4
Range	A group of cells	A2:D20
Row	Horizontal record	Row 5 = one customer
Column	One field/category	Column C = Sales

Example: a sales table

A (Date)	B (Product)	C (Quantity)	D (Sales)
01-Jan	Laptop	2	90000
02-Jan	Mouse	5	2500
03-Jan	Keyboard	3	4500

Practice: create the same table in Excel. Convert it to a Table with Ctrl + T.

1.2. Professional Data Structure

Good Excel analysis starts with clean structure.

A good raw dataset usually follows four rules:

- One header row. • One record per row. • One field per column.
- No merged cells, blank separator rows or decorative totals inside the raw data.

Bad structure vs good structure

Bad	Why it causes problems
Merged title above every section	Filters and PivotTables may not behave cleanly
Blank row after every 10 records	Breaks continuous data regions
Total row inside raw data	Can be double-counted in summaries and calculations
Jan, Feb, Mar as separate columns	Harder to analyze by date or group into quarters

Good structure

Date	Region	Product	Sales
01-Jan	North	Laptop	45000
02-Jan	South	Mouse	500
03-Jan	North	Laptop	45000

Why this matters: the same structure works for filters, SUMIFS, XLOOKUP, PivotTables & charts.

2. Cell References and Formula Basics

Learn how Excel moves references when formulas are copied.

Relative reference: =B2+C2 -> Excel returns 30. Becomes =B3+C3 when dragged down.

Absolute reference: =B2*\$H\$1 -> Result Rs.1,800. \$H\$1 remains strictly locked.

Mixed reference: \$A2 locks column A, while A\$2 locks row 2.

Quantity (A)	Price (B)	Revenue (C)
2	500	?
5	200	?
10	100	?

In C2 type =A2*B2 and drag down. Then change B3 and watch C3 update dynamically.

2.2. Core Arithmetic Functions

Write these without searching: SUM, AVERAGE, MIN, MAX, COUNT, COUNTA, ROUND.

Function	Formula	Example result
SUM	=SUM(B2:B5)	Total of B2:B5
AVERAGE	=AVERAGE(B2:B5)	Arithmetic average
MIN	=MIN(B2:B5)	Smallest number
MAX	=MAX(B2:B5)	Largest number
COUNT	=COUNT(B2:B5)	Numeric count
COUNTA	=COUNTA(A2:A5)	Non-empty cell count
ROUND	=ROUND(B2,2)	2 decimal places

Practice: replace four values with your own numbers and calculate total, average, min & max.

3. Text Functions and Data Cleaning

Exported business data is often inconsistent.

Contains	Formula	Result
john doe	=TRIM(A2)	john doe
john doe	=PROPER(A2)	John Doe
john doe	=PROPER(TRIM(A2))	John Doe
INV-89345-NY	=LEFT(A2,3)	INV
INV-89345-NY	=RIGHT(A2,2)	NY
INV-89345-NY	=MID(A2,5,5)	89345

Remember: MID requires text, starting position, and number of characters.

3.3. Joining and Measuring Text

Combine fields and inspect text length with precision.

Function	Formula	Use
LEN	=LEN(A2)	Count characters
CONCAT	=CONCAT(A2," ",B2)	Join fields
TEXTJOIN	=TEXTJOIN(" ",TRUE,A2:C2)	Join range with delimiter
SUBSTITUTE	=SUBSTITUTE(A2,"-","/")	Replace text
LOWER / UPPER	=LOWER(A2)	Case transformation

Company email: =LOWER(CONCAT(A2,".",B2,"@company.com")) -> arman.khan@company.com

4. Conditional Calculations

SUMIF and SUMIFS as key workplace calculation patterns.

Pattern	Syntax	Meaning
SUMIF (1 condition)	=SUMIF(A2:A5,"North",B2:B5)	Add Sales where Region = North
SUMIFS (Multiple)	=SUMIFS(C2:C100,A2:A100,"North",B2:B100,"Laptop")	Add Sales where Region=North AND Product=Laptop

Practice: create a 20-row sales table with Region, Product and Sales. Calculate North Laptop sales.

4.3. COUNTIF, COUNTIFS and AVERAGEIFS

Condition-based thinking for counts and averages.

Question	Formula
How many Approved?	=COUNTIF(C:C,"Approved")
How many Approved in North?	=COUNTIFS(A:A,"North",C:C,"Approved")
Average sales in North?	=AVERAGEIF(A:A,"North",D:D)
Average Laptop sales in North?	=AVERAGEIFS(D:D,A:A,"North",B:B,"Laptop")

Practice: calculate the number of Pending orders and the average approved Laptop sale.

5. IF, AND, OR and IFERROR

Logical decision branches in spreadsheets.

Function	Syntax	Example
IF	=IF(condition, true, false)	=IF(B2>=50000,"Target Met","Below Target")
AND	=AND(cond1, cond2)	=AND(B2="North",C2="Laptop")
OR	=OR(cond1, cond2)	=OR(B2="North",B2="South")
IFERROR	=IFERROR(val, fallback)	=IFERROR(XLOOKUP(A2,IDs,Prices),"Not Found")

Practice: create a Target Status column and a lookup result showing "Not Found" instead of error.

6. XLOOKUP: Connect Two Sheets

The modern lookup standard replacing VLOOKUP.

Sheet	Formula	Result
Master Sheet	Employee Master: E101..E104	Salary data
Report Sheet	=XLOOKUP(A2,Master!A2:A5,Master!B2:B5,"Not Found")	58,000 for E103, "Not Found" for E105
Absolute Lock	=XLOOKUP(A2,Master!\$A\$2:\$A\$5,Master!\$B\$2:\$B\$5,"Not Found")	Master range stays locked

Practice: add Department and Name columns to master and retrieve all three fields in report.

6.2. VLOOKUP and Troubleshooting

Understand VLOOKUP for existing legacy enterprise workbooks.

Problem	Symptom	Workplace Fix
Extra spaces	#N/A	TRIM the lookup key
Text vs Number	#N/A	Make both sides match format
Wrong column index	Wrong result	Verify column counting (1, 2, 3...)
Copied formula shifts range	Results change	Use \$ absolute references

Interview Question: Why is XLOOKUP more flexible? Searches in any direction, built-in Not Found.

7. Dates and Time

Master EOMONTH, EDATE, NETWORKDAYS and DATEDIF.

Need	Formula	Example
Today	=TODAY()	Current dynamic date
Month end	=EOMONTH(A2,1)	Last day of next month
Same date next month	=EDATE(A2,1)	15-Jan -> 15-Feb
Working days	=NETWORKDAYS(A2,B2)	Excludes Saturday and Sunday
Tenure	=DATEDIF(A2,TODAY(),"Y")	Full elapsed years

Practice: calculate due date, working days and employee tenure for 10 records.

8. Conditional Formatting

Make critical trends visible without manual coloring.

Rule Type	Formula / Setting	Key Rule
Cell highlight	Greater Than -> 50000	Automates top value highlight
Whole-row highlight	=D2="Complete"	Lock column \$D, let row evaluate row-by-row
Common mistake	=D\$2="Complete"	Locks evaluation to D2 only, breaking row logic

Practice: highlight overdue dates, duplicate employee IDs, and rows where Priority = "High".

8.2. Data Validation and Drop-downs

Prevent typos and dirty inputs at data capture.

Step	Action	Benefit
1	Select Status cells	Target range defined
2	Data -> Data Validation	Configure rule
3	Allow: List -> Source: Approved, Pending, Rejected	Prevents typing mistakes and inconsistent filters

Practice: create Status, Priority and Department drop-downs with warning alerts on mismatch.

9. PivotTables: Learn by Building One

The most powerful workplace reporting and summarization tool.

Step	Action	Pivot Field Destination
1	Click inside dataset	Active region
2	Insert -> PivotTable	New sheet / Range
3	Region	Rows area
4	Sales	Values area (Sum)
5	Product	Columns area

Practice: add 20 more rows. Create a second PivotTable showing average sales by region.

9.2. PivotTables: Slicers, Dates & Refresh

Move from a simple summary to an interactive report.

Feature	Action	Workplace Advantage
Slicers	Insert -> Slicer -> Category	Visual 1-click filtering buttons
Date grouping	Right click date -> Group -> Months	Automatic monthly/quarterly trends
Refresh	Right click -> Refresh	Excel Table source auto-expands range

Practice Project: 500 sales rows, monthly revenue by store, category slicer, and line PivotChart.

10. Charts: Choose the Right Visual

Charts must answer a question, not simply decorate.

Question	Good visual choice	Example
Which store sold more?	Column / Bar	Sales comparison by store
How did sales change over time?	Line	Monthly sales trend
How are two measures related?	Scatter	Price vs quantity correlation
What share comes from each?	Doughnut	Category product mix
Compare revenue and margin %?	Combo (Column + Line)	Revenue (bars) + Margin (secondary line)

Checklist: clear title, stated units, readable category labels, no 3D distortion, max 4 series.

11. Power Query: Manual Work to Refresh

Connect to external data, clean step-by-step, and refresh.

Step	Power Query Action	Result
1-2	Data -> Get Data -> From Folder	Points to reports folder
3-5	Remove unnecessary columns, change types	Ensures clean schema
6-7	Clean text and remove unwanted blanks	Standardized rows
8	Close & Load -> Refresh All	Automates weekly report forever

Practice: create 3 small CSV files with matching headers. Combine them using From Folder.

11.2. Power Query: Merge vs Append

The fundamental architectural distinction in data modeling.

Operation	Think of it as	Example
Merge	Join columns horizontally	Orders + Customer Master via Customer ID
Append	Stack rows vertically	January + February + March sales into 1 table

Practice: create two customer files and three monthly sales files. Perform one Merge and one Append.

12. Dashboard Building

A summary layer engineered for executive decision-makers.

Section	Placement	Content
Top	KPI Row	Total Sales Orders Avg Order Value Growth %
Middle Left	Trend Zone	Monthly Sales Trend (Line Chart)
Middle Right	Breakdown Zone	Sales by Region (Bar Chart)
Bottom	Detail Zone	Top Products / Exceptions
Side / Top	Filter Zone	Interactive Slicers

Build order: Clean data -> Pivots -> KPIs -> Charts -> Slicers -> Align & Format -> Test.

13. Modern Excel Functions

Dynamic arrays that spill results automatically into adjacent cells.

Function	What it does	Simple example
UNIQUE	Returns distinct values	=UNIQUE(B2:B100)
FILTER	Returns matching rows	=FILTER(A2:D100,D2:D100="Pending")
SORT	Sorts an array	=SORT(A2:D100,4,-1)
SORTBY	Sorts by another range	=SORTBY(A2:D100,D2:D100,-1)
SEQUENCE	Generates numbers	=SEQUENCE(10)
LET	Names intermediate values	=LET(x,B2*C2,x*18%)

FILTER worked example: =FILTER(A2:D100,D2:D100="Pending") spills matching rows automatically.

14. Practical Error Handling

Do not hide errors blindly. Diagnose why they occurred.

Error	Meaning	Workplace Fix
#N/A	No matching lookup value	Check lookup key, extra spaces, lookup range
#VALUE!	Wrong value / data type	Check text inside arithmetic calculations
#REF!	Broken reference	Repair deleted referenced cell, row or column
#DIV/0!	Division by zero	Check denominator or wrap with IF/IFERROR
#NAME?	Unrecognized formula name	Check spelling of function and quotes around text

Example: IFERROR(XLOOKUP(A2,Master!A:A,Master!B:B),"Not Found"). Do not hide bad master data.

15. Job-Ready Project 1: HR Roster

Data cleaning, corporate emails, joining alerts and validation.

Task #	Deliverable	Production Formula
1	Convert names to Proper Case	=PROPER(TRIM(A2))
2	Create corporate emails	=LOWER(SUBSTITUTE(A2," ",".")&"@company.com")
3	Upcoming joins (next 7 days)	=IF(AND(B2>=TODAY(),B2<=TODAY()+7),"Joining Soon","Later")
4	Conditional formatting	Highlight records flagged as "Joining Soon"
5	Data validation	Department drop-down for future entries

Practice result: build the entire sheet from scratch with at least 30 employee records.

15.2. Job-Ready Project 2: Inventory Match

Practice multi-sheet XLOOKUP and conditional price variance flagging.

Component	Formula / Logic	Outcome
Warehouse sheet	Product ID, Old Price, New Price	Target tracking table
Vendor sheet	Product ID, New Price	Source catalog
Price lookup	=XLOOKUP(A2, Vendor!A:A, Vendor!B:B, "Not Found")	Retrieves updated prices
Price variance	=IF(C2>B2, "Price Increased!", IF(C2<B2, "Price Decreased", "No Change"))	Flags margin impact

Practice: create 50 products, including at least 5 products that do not exist in the vendor list.

15.3. Job-Ready Project 3: Monthly Sales MIS

Flagship portfolio project combining tables, pivots, slicers and KPIs.

Step	Deliverable	Key Technique
1	Convert data to Table	Ctrl + T structured references
2	Clean text fields	TRIM and PROPER
3	PivotTable for monthly revenue	Group dates by month, store in rows
4	Product Category slicer	Interactive filter button
5	Line PivotChart	Shows monthly trajectory
6	KPI Cards	Total Sales and Average Order Value
7	Executive observations	Write 3 observations from the report

This is the project you should be able to walk through and explain in an interview.

16. Excel Interview Preparation

Prepare for concept questions and timed practical assessments.

Question	Core Concept	Executive Answer Summary
1	Relative vs Absolute	\$ locks column and/or row when dragging formula
2	XLOOKUP vs VLOOKUP	Searches in any direction, immune to column shifts
3	SUMIF vs SUMIFS	Single vs multiple criteria; sum_range order difference
4	Merged cells hazard	Breaks filters, formulas, sorting and PivotTables
5	Merge vs Append	Merge joins columns; Append stacks rows

Timed Practical Test: given 2,000 rows, clean data, join prices, pivot, slice and chart in 40 mins.

17. 30-Day Learning Roadmap

Structured daily progression from basics to job-ready analyst.

Days	Study Focus	Hands-on Task
1-4	Interface, Tables, References, Arithmetic	Build sales calculator and clean table
5-10	IF, Text Cleaning, Flash Fill	Clean dirty customer list and format emails
11-16	SUMIFS, COUNTIFS, XLOOKUP, VLOOKUP	Regional sales analysis and inventory match
17-20	Dates, Validation, Formatting	Due-date tracker and HR roster
21-25	PivotTables, Slicers, Charts	Sales MIS and management presentation
26-28	Power Query Automations	Multi-CSV folder consolidation
29-30	Capstone Projects & Mock Tests	Timed practical interview test

Daily rule: 30% time learning concepts, 70% actually typing, breaking and fixing spreadsheets.

18. Final Job-Ready Checklist

Comprehensive checklist before applying for Excel-heavy analyst roles.

#	Skill Competency	Status
1	Create clean Excel Tables (Ctrl+T) from raw exported data	Verified
2	Understand relative, absolute (\$) and mixed references	Verified
3	Clean text with TRIM, PROPER, LEFT, RIGHT, MID, SUBSTITUTE	Verified
4	Multi-criteria formulas with SUMIFS, COUNTIFS, AVERAGEIFS	Verified
5	Connect sheets with XLOOKUP and troubleshoot VLOOKUP	Verified
6	Calculate dates, month-ends (EOMONTH) and working days	Verified
7	Build and refresh PivotTables, Slicers and PivotCharts	Verified
8	Power Query Merge and Append automation	Verified
9	Build executive dashboards and troubleshoot errors	Verified
10	Explain your analytical approach in an interview	Verified

The finish line: You are job-ready when you can reason about unfamiliar data and choose the right tool.

Excel Quick Revision Sheet

Fast reminder for interviews, practical tests and daily desk reference.

Task Need	Start With	Core Syntax
Add numbers	SUM	=SUM(B2:B100)
Conditional total	SUMIFS	=SUMIFS(D:D,A:A,"North",B:B,"Laptop")
Apply rule	IF / AND / OR	=IF(B2>=50000,"Target Met","Below Target")
Safe lookup	XLOOKUP	=XLOOKUP(A2,Master!A:A,Master!B:B,"Not Found")
Clean name	TRIM / PROPER	=PROPER(TRIM(A2))
Extract text	LEFT / RIGHT / MID	=MID(A2,5,5)
Working days	NETWORKDAYS	=NETWORKDAYS(A2,B2)
Month end date	EOMONTH	=EOMONTH(A2,1)
Whole row highlight	Condition	=\$D2="Complete"

Core Shortcuts: Ctrl+T Table | Ctrl+Shift+L Filter | Ctrl+E Flash Fill | Ctrl+1 Format | Alt+= Sum

Core Shortcuts & Final Workflow

Speed shortcuts and the universal workplace analytical methodology.

Key Combo	Action	Usage Context
F2	Edit active cell	In-place formula editing
Ctrl + Arrow	Jump to edge of data region	Fast navigation across large sheets
Ctrl + Shift + Arrow	Select to edge of data region	Instant column/row selection
Ctrl + ;	Insert current date	Static timestamping
Ctrl + Shift + ;	Insert current time	Static time recording
Alt + H, O, I	AutoFit column width	Cleans clipped column display

5-Step Universal Sequence: Clean -> Validate -> Calculate -> Lookup -> Summarize -> Visualize -> Check.